The Scale of the Universe

Exam code: 0625 & 0972

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The Milky Way

The Universe

- The Universe is defined as

A large collection of billions of galaxies

- It is also the name given to the entirety of space

Galaxies

- A galaxy is defined as

A large collection of billions of stars

- Stars are large astronomical objects, such as the Sun

The Milky Way

- **The Milky Way** is one of many billions of galaxies making up the Universe
- **The Sun** is one of many billions of stars making up the Milky Way
- Other stars in the Milky Way galaxy are **much further away** from Earth than the Sun is
- Some of these stars also have **planets** which orbit them

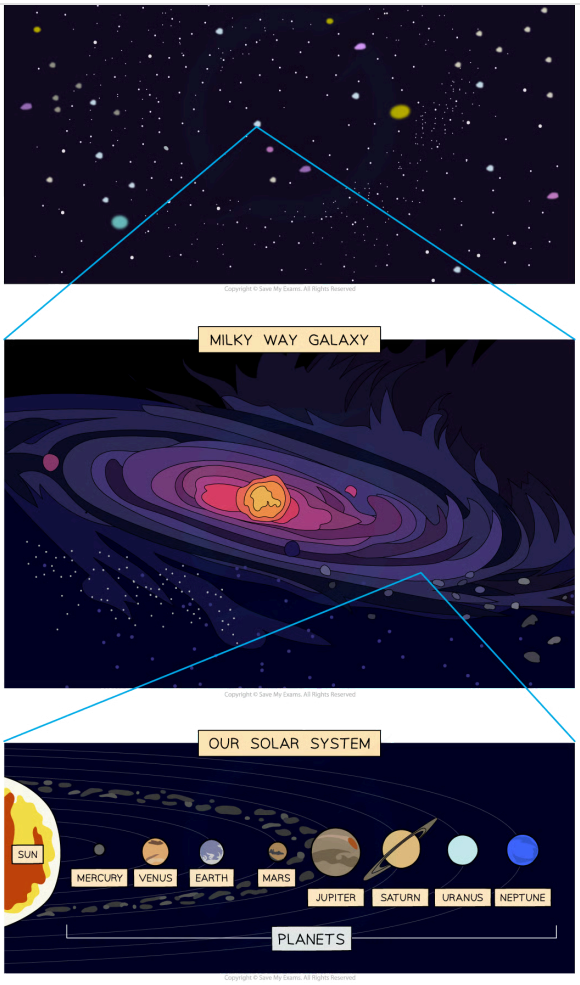
Hierarchy of the Universe

Help me understand

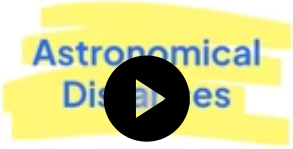
Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



The Universe is a large collection of galaxies and a galaxy is a large collection of stars. The Sun is a star at the centre of our Solar System in the Milky Way galaxy



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Astronomical distances

unit called the **light year**

- One light-year is defined as:

The distance travelled by light in one year

- The **diameter of the Milky Way** is approximately **100 000 light-years**
- This means that light would take 100 000 years to travel from one side of the Milky Way to the other

Extended tier only

- One light year is equal to 9.5×10^{12} km, or $\mathbf{9.5 \times 10^{15}}$ **m**

Worked Example

The centre of our galaxy is 30 000 light years away.

(a) How long does it take light to reach the Earth?

Extended tier only

(b) Calculate this distance in km.

Answer:

(a)

- The centre of our galaxy is 30 000 light years away
- It takes light 30 000 years to reach the Earth from the centre of our galaxy

(b)

Extended tier only

Step 1: Write down the known quantities:

- The centre of our galaxy is 30 000 light years away
- 1 light year = 9.5×10^{12} km

Step 2: Calculate the distance in km:

$$\text{distance} = 30\,000 \times (9.5 \times 10^{12})$$

$$\text{distance} = 2.85 \times 10^{17} \text{ km}$$

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